MicroSD Card Subsystem & SPI Logging Driver

1. Overview

The **MicroSD Card Subsystem** provides non-volatile offline persistence for the Smart Pill Box. It serves two critical operational roles:

1. **Intake Event History Logging (`LOGS.CSV`)**: Creates an audit trail recording exact date, time, event types (dose taken, tamper / wrong pill attempt, clock adjustments), and verification notes.

2. **Persistent Configuration (`CONFIG.TXT`)**: Stores user-configured Pill 1 and Pill 2 alarm times so schedules survive full power outages, battery disconnects, or MCU resets.

---

2. Hardware Architecture & Pin Allocation (SPI1)

The MicroSD module interfaces with the STM32F103C8T6 via **hardware Serial Peripheral Interface (SPI1)** on Port A, leaving Port B entirely free for buttons, alerts, and I2C:

SD Card Module Pin	STM32 Blue Pill Pin	Mode / Function	Electrical Details
CS (Chip Select)	**`PA4`**	GPIO Output (Push-Pull)	Active LOW card select
SCK (Clock)	**`PA5`**	Alternate Function (**`SPI1_SCK`**)	Generated SPI clock
MISO (Master In)	**`PA6`**	GPIO Input with Internal Pull-Up (**`SPI1_MISO`**)	Serial data out from SD
MOSI (Master Out)	**`PA7`**	Alternate Function (**`SPI1_MOSI`**)	Serial data in to SD
VCC	**`5V`** (or **`3.3V`**)	Power Supply	5V if module has onboard 3.3V regulator; 3.3V for bare breakout
GND	**`GND`**	Ground Reference	Common ground with STM32

---

3. Communication Protocol: SPI Mode

Although SD cards natively communicate using the 4-bit parallel SD Bus mode (SDIO), the STM32F103C8T6 utilizes **SPI mode** due to its low pin-count requirement and deterministic timing.

3.1 Card Initialization Sequence

When entering SPI mode upon power-up, the firmware executes the official SD Association card initialization flow:

```
flowchart TD
    A[Power ON] --> B[Drive CS HIGH and send 80+ dummy clock cycles]
    B --> C[Send CMD0: GO_IDLE_STATE with CRC 0x95]
    C -->|Response 0x01: Idle| D[Send CMD8: SEND_IF_COND 0x1AA with CRC 0x87]
    C -->|No Response| ERR[Flag SD Unavailable & Fallback]
    D -->|Echo 0x1AA Valid| E[Loop CMD55 + ACMD41: SD_SEND_OP_COND]
    E -->|Response 0x00: Ready| F[Switch SPI Clock to High Speed 9 MHz]
    E -->|Timeout| ERR
    F --> G[Card Ready: Mount File System]
```

1. **Dummy Clock Train**: SD cards start in native SD bus mode. Clocking at least 74 cycles with **`CS = HIGH`** instructs the internal card controller to prepare for SPI negotiation.

2. **`CMD0` (Software Reset)**: Forces the card into SPI mode. Response must be **`0x01`** (**`In Idle State`**).

3. **`CMD8` (Voltage & Interface Condition)**: Inquires if the card supports SD Version 2.0+ (voltage range $2.7\text{ V} - 3.6\text{ V}$).

4. **`CMD55` + `ACMD41` (Operating Condition)**: Loops until the card completes its internal initialization and returns **`0x00`**.

5. **High-Speed Switchover**: Initial identification occurs below 400 kHz (**`SPI1`** Prescaler **`/256`** @ 72 MHz $\approx 281\text{ kHz}$). Once initialized, clock prescaler is increased to **`/8`** (9 MHz) for rapid file read/write operations.

---

4. File Formats & Persistence Engine

4.1 Intake History Log (`LOGS.CSV`)

Every critical patient interaction is captured in standard comma-separated values format, openable in Microsoft Excel, LibreOffice, or Google Sheets:

```
DATE,TIME,EVENT,DETAILS
2026-11-02,07:30:12,DOSE_1_TAKEN,Confirmed Button 1
2026-11-02,19:00:45,WRONG_BUTTON,Pill 2 due - Pressed Btn 1
2026-11-02,19:01:05,DOSE_2_TAKEN,Confirmed Button 2
2026-11-02,21:15:00,CLOCK_ADJUST,RTC clock re-calibrated
2026-11-02,21:16:10,CONFIG_UPDATE,Pill 1 time updated
```

4.2 Configuration File (`CONFIG.TXT`)

On boot, **`SD_LoadConfig()`** reads alarm hours and minutes:

```
PILL1_HOUR=07
PILL1_MIN=30
PILL2_HOUR=19
PILL2_MIN=00
```

Whenever the user updates alarm times via the 3-button setup menu, **`SD_SaveConfig()`** immediately rewrites this file.

---

5. Fault-Tolerance & Hot-Plug Handling

Embedded medical devices must never lock up due to missing or removed accessories:

- **Non-Blocking Fallback**: If an SD card is omitted or detached, **`SD_Init()`** fails cleanly, sets **`sdAvailable = false`**, and allows the pill box to run normally.
- **Live Status Reporting**: The OLED boot screen displays:

```
  I2C BUS SCAN: 0x3C 0x68
  RTC: OK    SD: OK
```

If unmounted, it reports **`SD: NONE`** without blocking execution.

---

6. Potential Evaluator Questions & Answers

- **Q: Why use SPI instead of SDIO for the MicroSD card?**
- *A: The STM32F103C8T6 (LQFP48) does not feature hardware SDIO peripheral blocks (SDIO is present only on higher-density F103Rx/Vx lines). SPI mode provides universal MicroSD compatibility using only 4 pins.*
- **Q: Why did you initialize SPI at low speed first before switching to high speed?**
- *A: The SD physical layer specification mandates that card initialization must occur at a clock frequency between 100 kHz and 400 kHz. After receiving the* `ACMD41` *ready response (*`0x00`*), the MCU safely scales the clock up to 9 MHz.*
- **Q: How are file operations protected against sudden power loss?**
- *A: Log writes flush immediately with verified block termination tokens and card deselection (*`SD_Deselect()`*), minimizing the unwritten buffer window to microseconds.*